

# Dynamic Response of Magnetic Ellipsoid Raspberry Dipoles

29.6.2021

Florian Kuderna

# Outline

1. Task Definition
2. Dynamic Susceptibility
3. Building Ellipsoidal Raspberries
4. Simulation (Parameters/Issues)
5. Results
6. Outlook

# Task

Build ellipsoidal raspberry particle with magnetic moment along the long axis

1. Build ~~ellipsoidal raspberry dipoles~~ in LB-Fluid
2. Measure dynamic response in the LB-fluid
3. Check whether results differ from spherical raspberries

# Magnetic Susceptibility $\chi$

- Determine the response of the magnetization  $M$  to an external field  $H$

$$\chi_{stat} = \left. \frac{\partial M}{\partial H} \right|_{H=0}$$

- If AC field, frequency dependent dynamic magnetic susceptibility:

$$\chi(\omega) = \chi'(\omega) + i\chi''(\omega)$$

- Power loss is proportional to imaginary part ( $\rightarrow$  Heat)  
useful eg. for medical applications

# Linear Response Theory

- Dynamic Susceptibility in terms of zero-field magnetization autocorrelation function:

$$\chi(\omega) = \chi'(\omega) + i\chi''(\omega) \rightarrow \chi(\omega) = \chi(0) \left( 1 + i\omega \int_0^\infty C(t) e^{i\omega t} dt \right)$$

with  
static susceptibility:

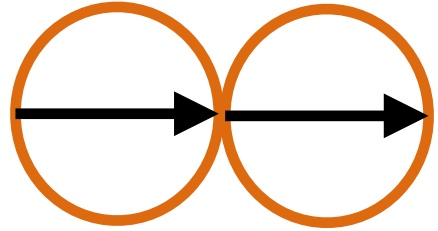
$$\chi(0) = \frac{\mu_0 \langle \vec{M}^2 \rangle}{3Vk_b T}$$

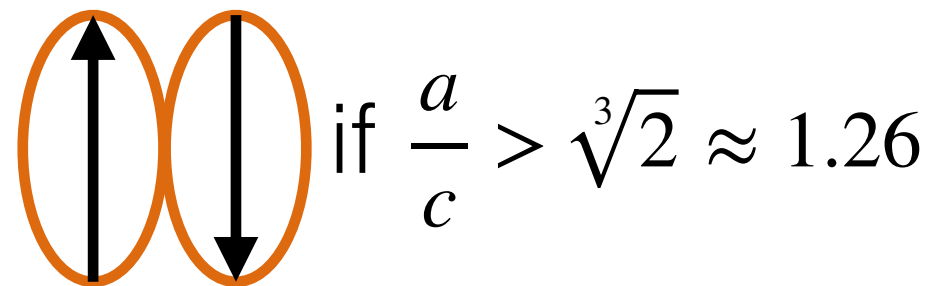
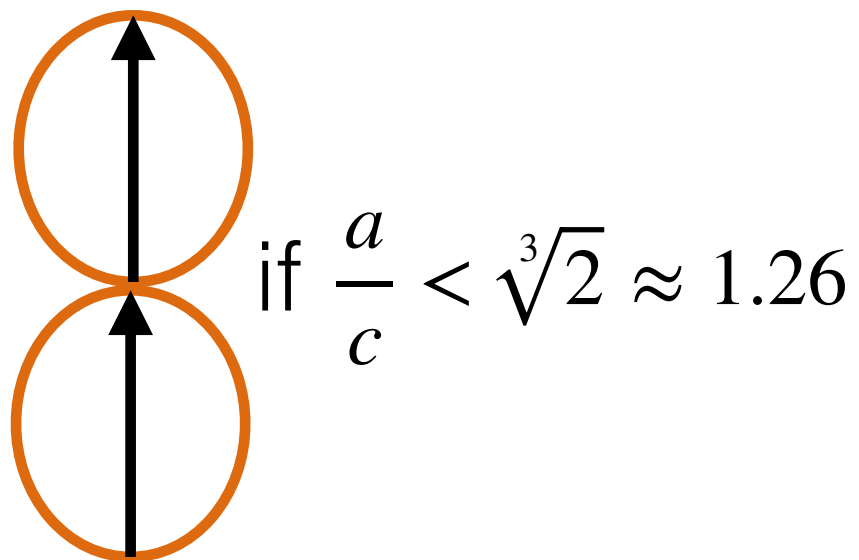
and the MACF:

$$C(t) = \frac{\langle \vec{M}(0) \cdot \vec{M}(t) \rangle}{\langle \vec{M}^2 \rangle}$$

$$\vec{M}(t) = \sum_{i=1}^N \vec{\mu}_i(t)$$

# Why Ellipsoids?

- Dipole-Dipole Interaction:  $U_d(ij) = -\frac{\mu_0}{4\pi} \left( 3 \frac{(\vec{m}_i \cdot \vec{r}_{ij})(\vec{m}_j \cdot \vec{r}_{ij})}{|\vec{r}_{ij}|^5} - \frac{(\vec{m}_i \cdot \vec{m}_j)}{|\vec{r}_{ij}|^3} \right)$
- Dipole Coupling Constant:  $\lambda = \frac{m^2}{k_B T \sigma^3}$
- Ground state for dipolar spheres: 
- Ground states for ellipsoids:



magnetic moment cancels  
 $\rightarrow$  suscept. decreases

# Why Raspberries?

- Raspberries have a shape
- Raspberries have surface structure
  - > Interaction with the fluid
  - > Hydrodynamic effects become relevant



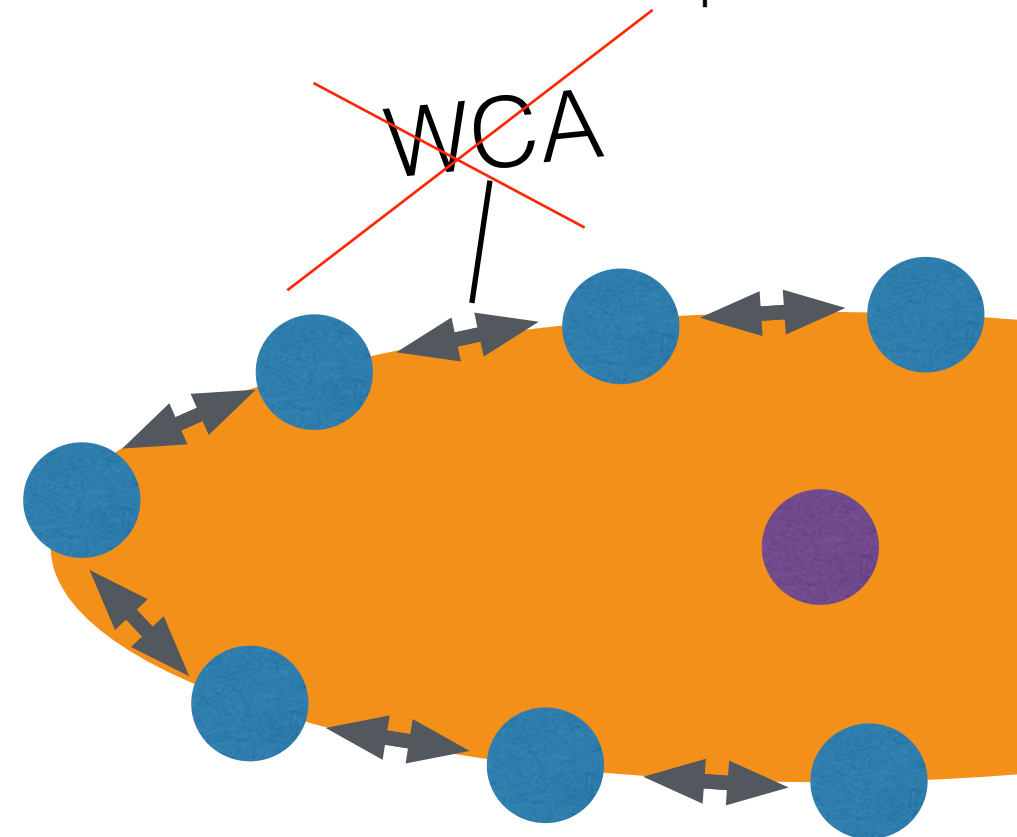
# Building Ellipsoidal Raspberries

# Outer Shell Spacing

WCA is used for beads from different raspberries

- Central Particle
- Place outer shell particles on ellipsoid around central particle
- Outer layer particles spaced homogeneously through repulsive WCA potential:

$$u_{ij}^{WCA} = \begin{cases} 4\epsilon \left[ \left( \frac{\sigma}{r_{ij}} \right)^{12} - \left( \frac{\sigma}{r_{ij}} \right)^6 \right] + \epsilon & r_{ij} \leq r_{min} \\ 0 & r_{ij} > r_{min} \end{cases}$$



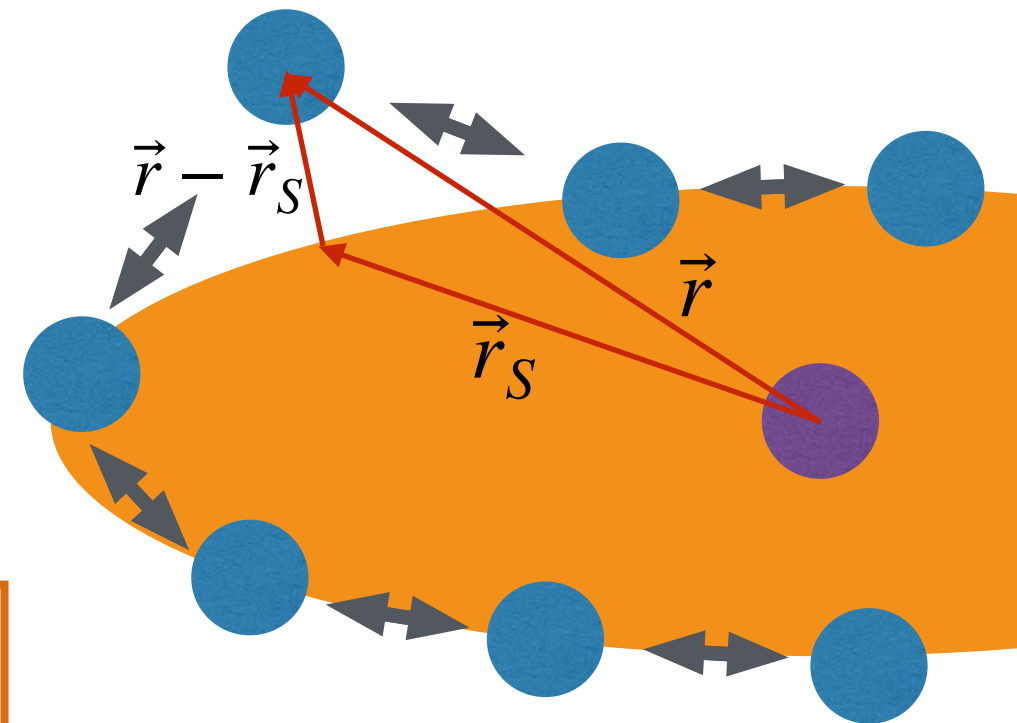
# Outer Shell Stability

- Shell particles get pushed off ellipsoid surface
- To place them back calculate nearest surface position for the particle:

$$x_s = \frac{x}{\left(\frac{t}{a^2} + 1\right)} \quad y_s = \frac{y}{\left(\frac{t}{b^2} + 1\right)} \quad z_s = \frac{z}{\left(\frac{t}{c^2} + 1\right)}$$

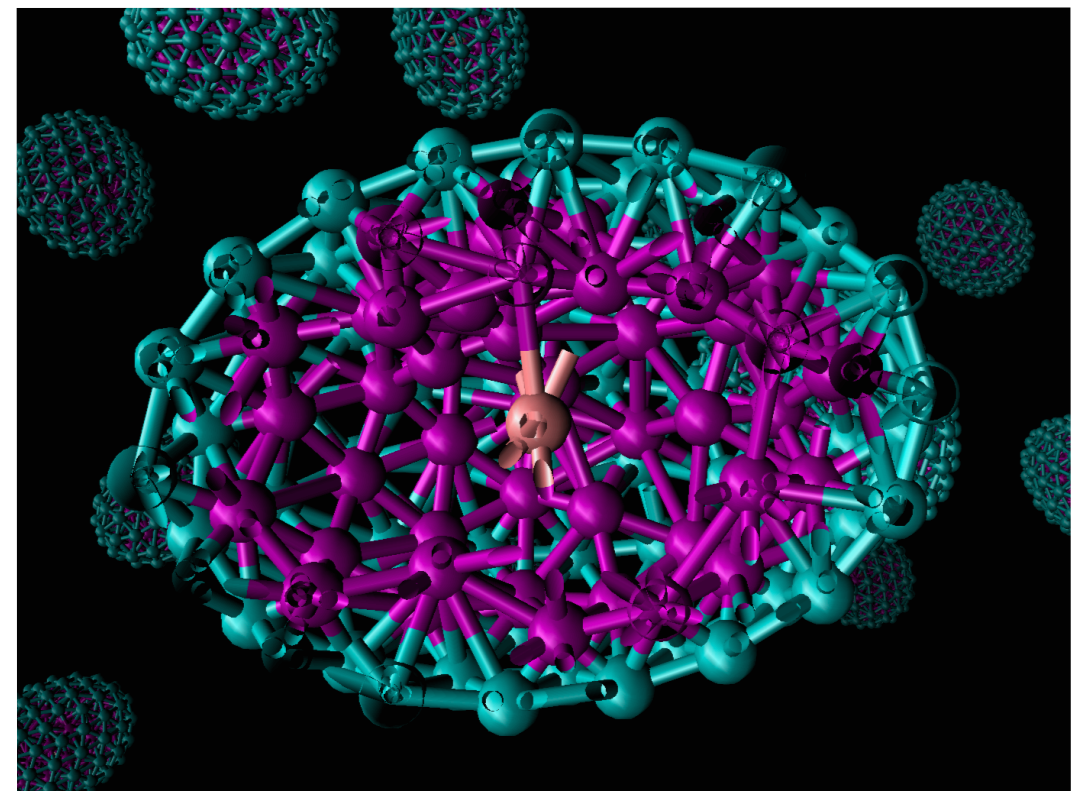
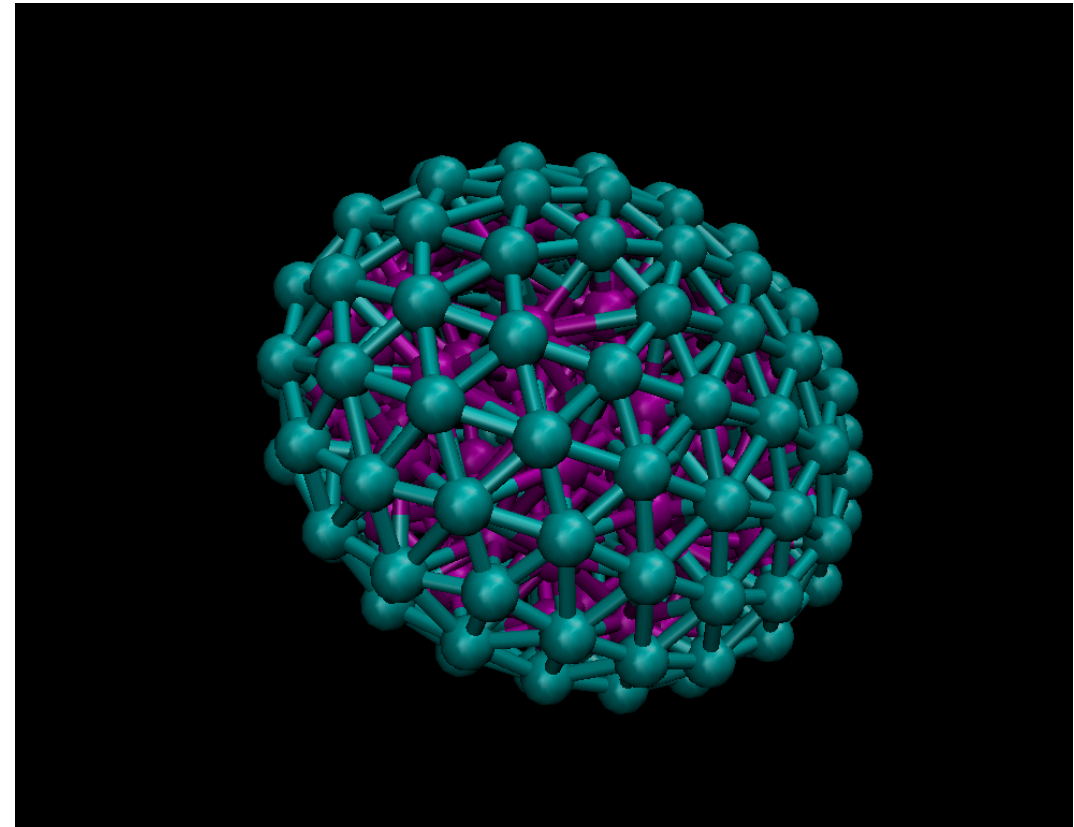
use largest  $t$  that gives a solution to:

$$F(t) = \left(\frac{xa}{t + a^2}\right)^2 + \left(\frac{yb}{t + b^2}\right)^2 + \left(\frac{zc}{t + c^2}\right)^2 - 1 = 0$$



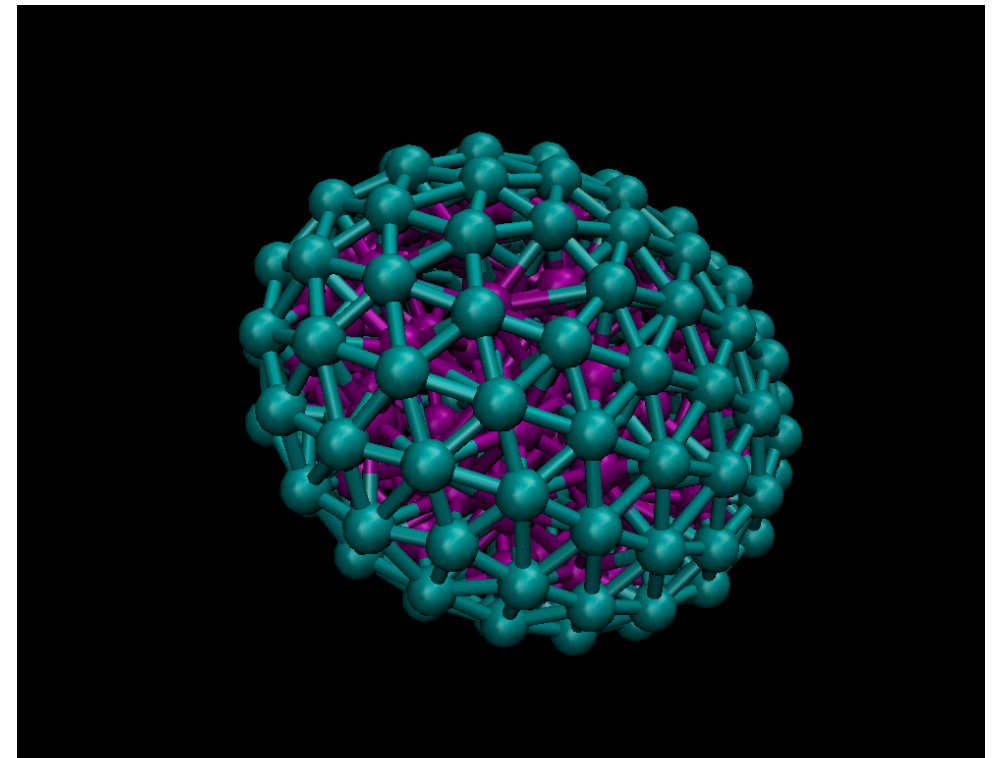
# Raspberry Filling

- Raspberry filled with particles acting as hydrodynamic coupling points for LB-Fluid
- Inner particles distributed on concentric layers
- Inner layers use WCA potential for spacing
- Inner layers use same procedure as outer layer to stay on the surface



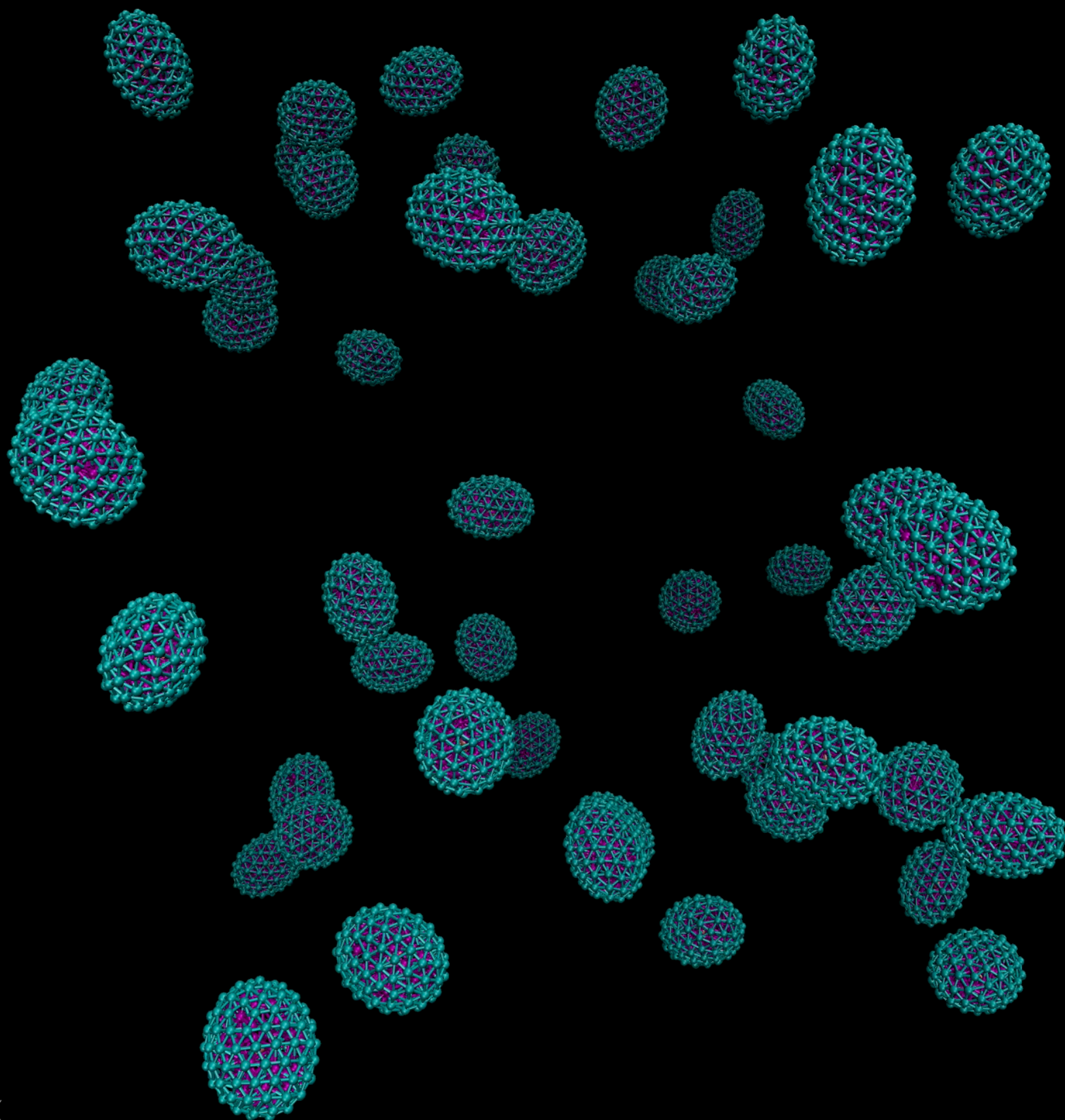
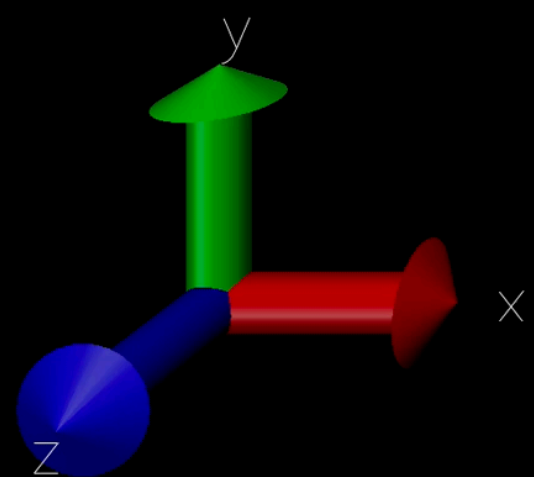
# Simulation Parameters

- Raspberry semi-axes:  
 $a = 3.0$  ;  $b = 2.5$  ;  $c = 2.0$
- $\lambda = 3$  &  $5$
- 50 Raspberries ( $m = 47.77$ )
- Volume fraction = 0.0125
- timestep = 0.001
- $3 \cdot 10^6$  integrations
- Simulation carried out with ESPResSo



LB-Fluid:

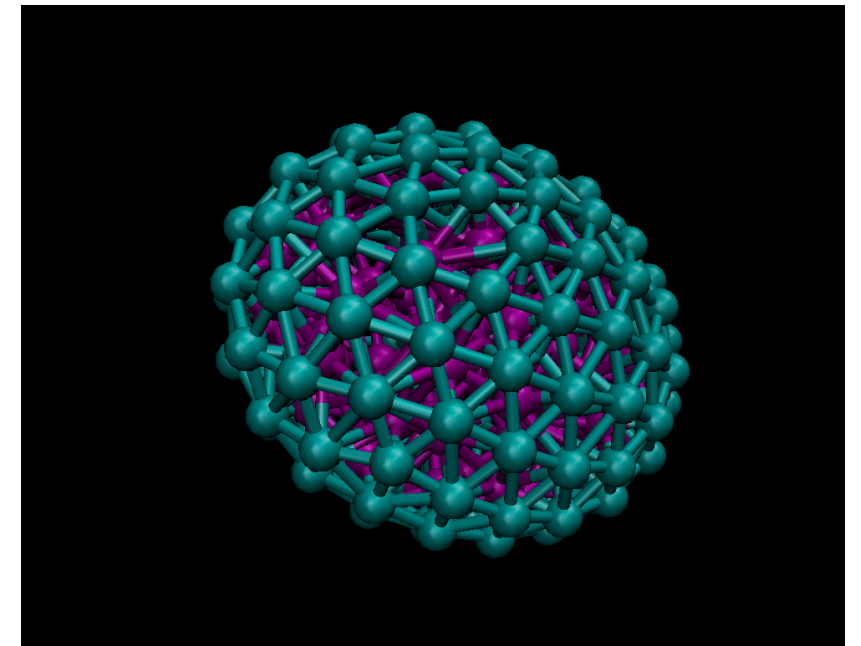
- grid = 1
- $kT = 1$
- density = 1
- viscosity = 3.39
- $\tau = 0.01$



# Issues and Expectations

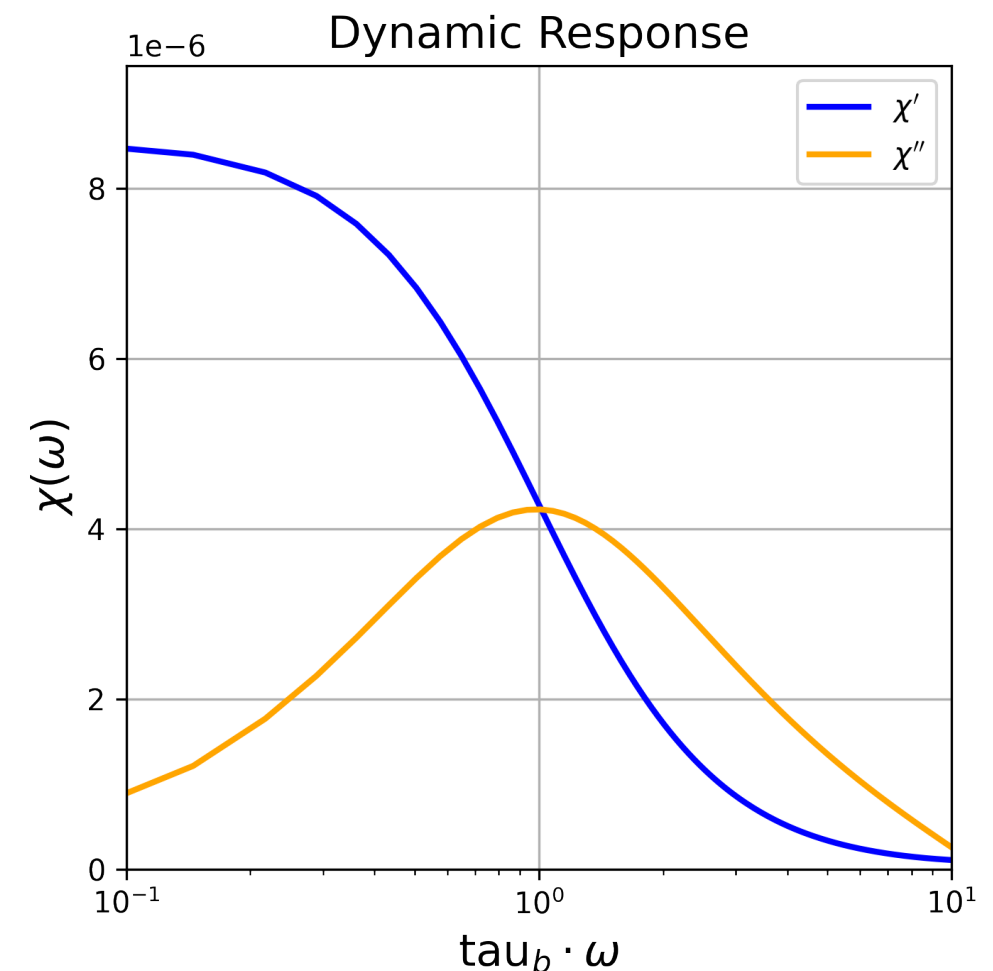
## Issues:

- Not enough simulation time
  - > MACF did not work
  - > Pair Formation low
  - > RDF inconclusive



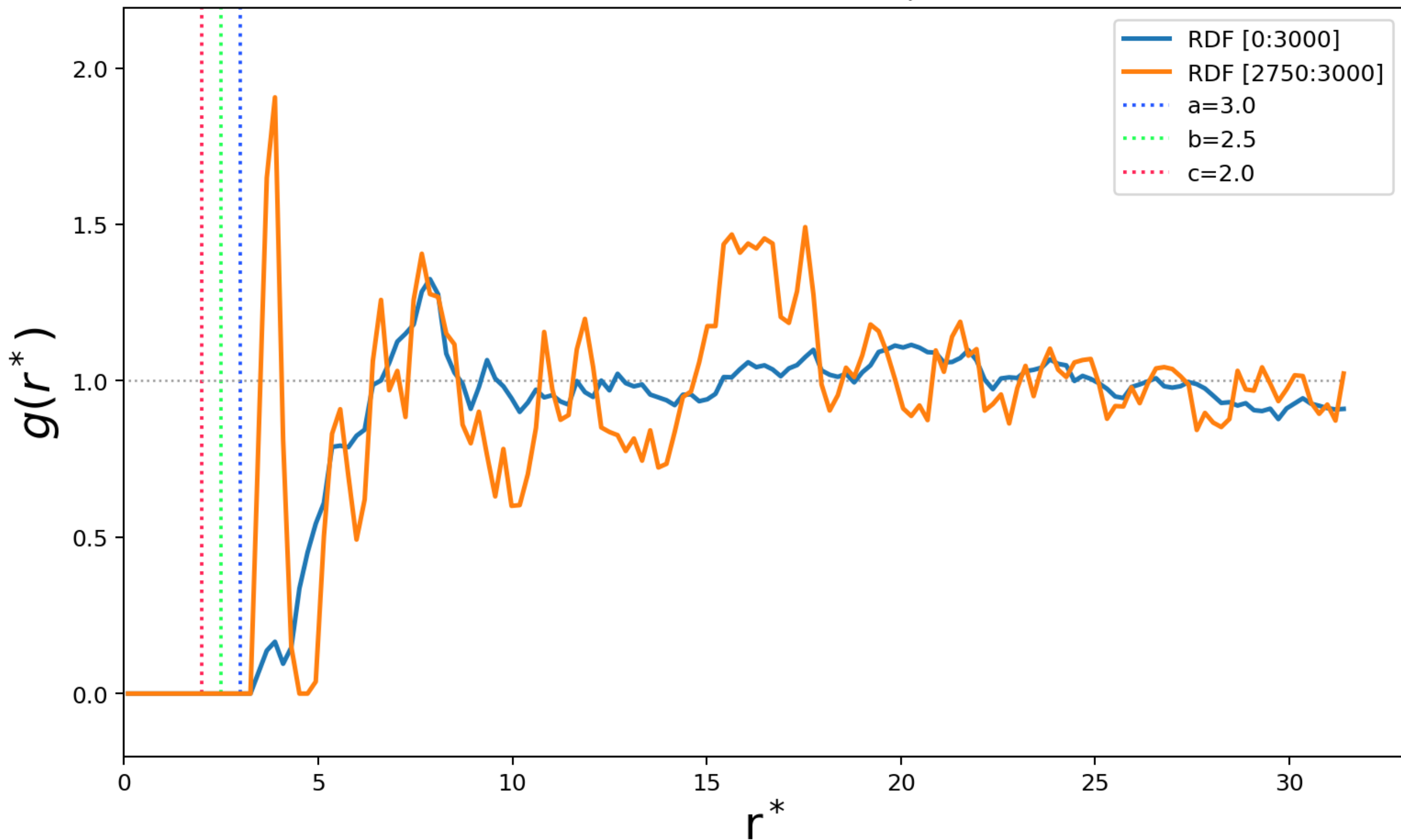
## Expectations if issues resolved:

- Lower static susceptibility for ellipsoidal particles
- $\text{Re}(\chi(\omega))$ : Vanishes above resonant frequency
- $\text{Im}(\chi(\omega))$ : Maximum at RF



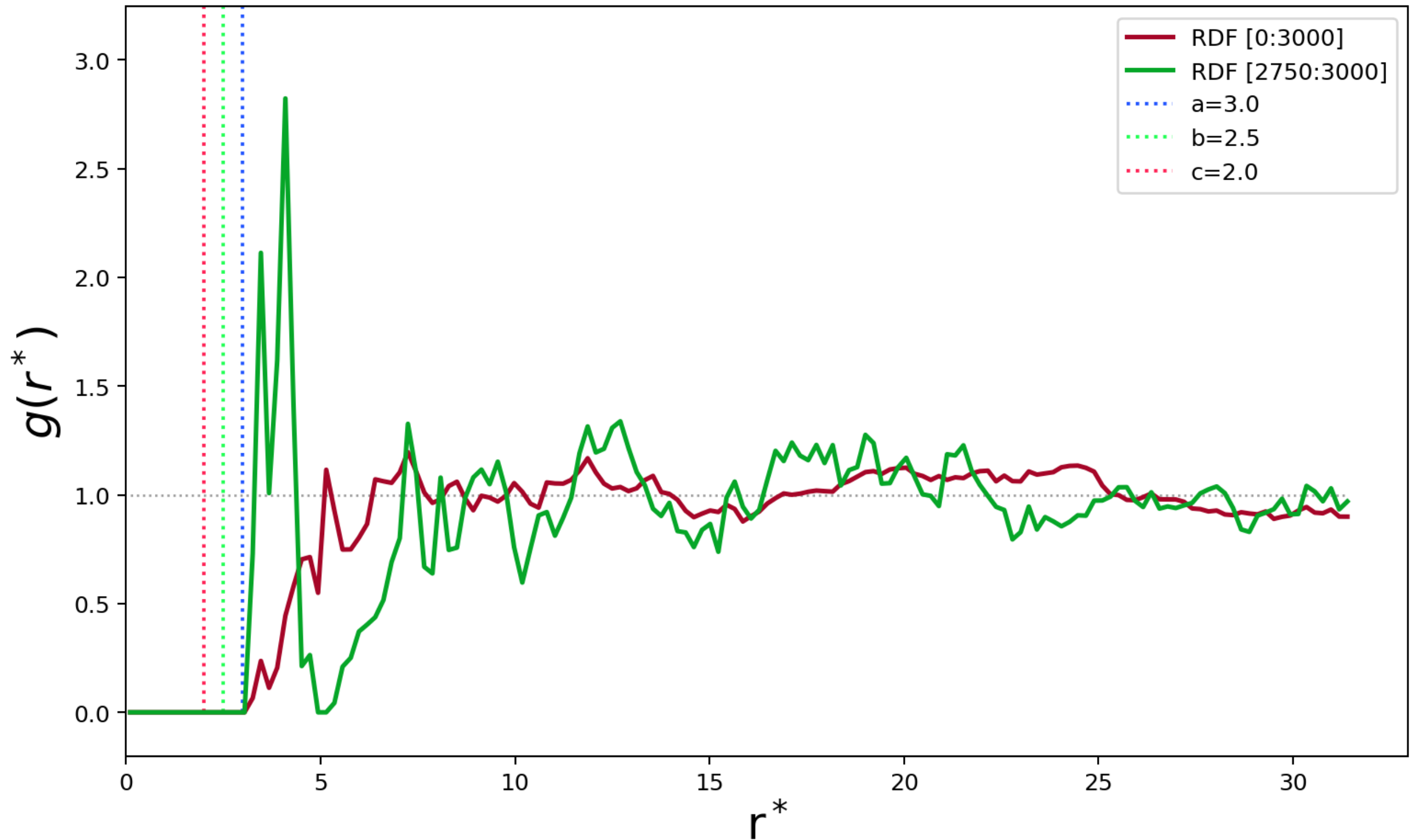
# RDFs

Radial Distribution Function  $\mu = 9.0$



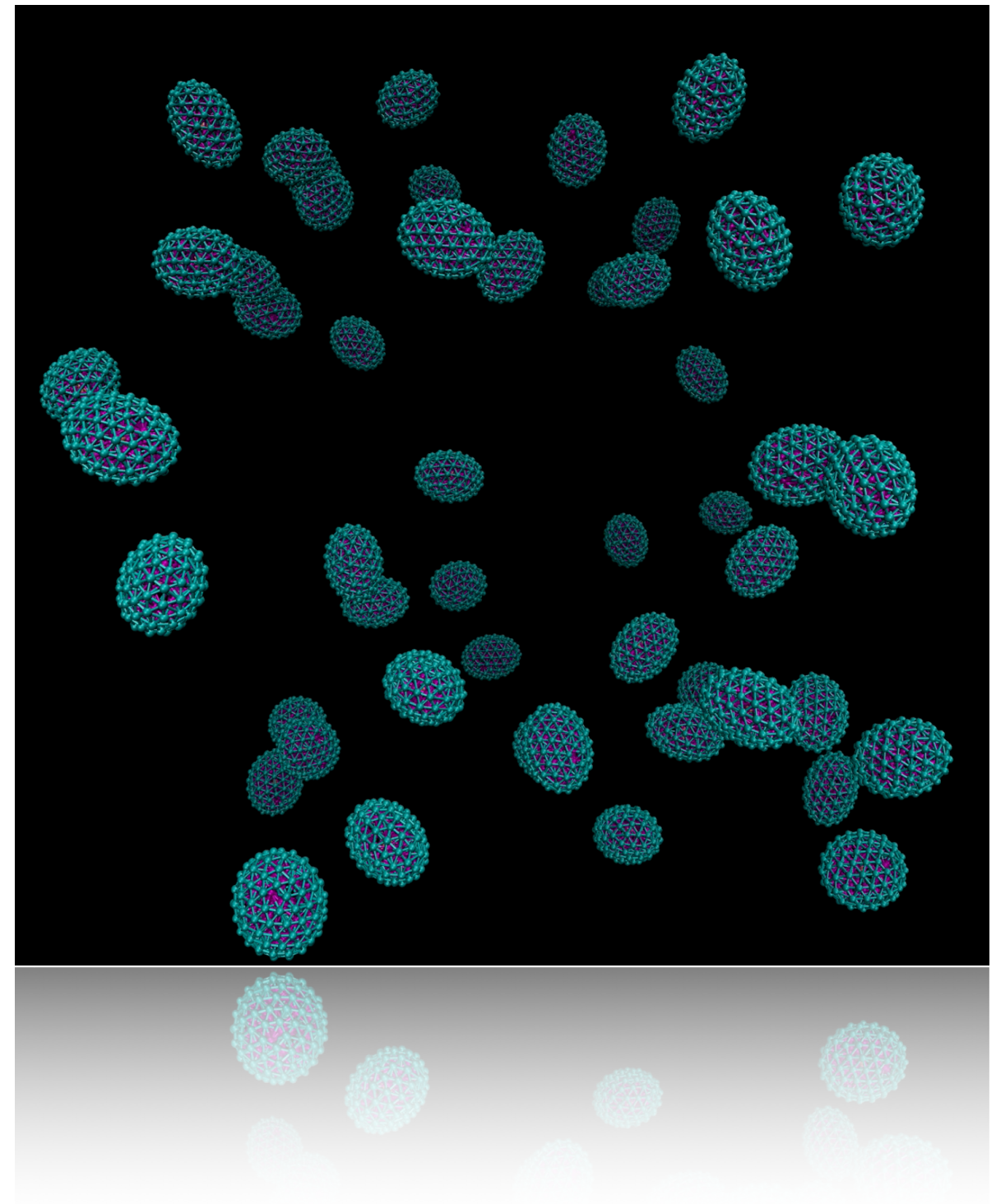
# RDFs

Radial Distribution Function  $\mu = 11.0$



# Outlook & Next Steps

- Longer Simulation Runs (10x)
- More Raspberries in System (5-10x)
- —> Retrieve MACF
- Measure Effect of Shape on Dynamic Susceptibility



# References

- V. D. Ustach and R. Faller. The raspberry model for protein-like particles: Ellipsoids and confinement in cylindrical pores. The European Physical Journal Special Topics, 225(8-9):1643–1662, Oct 2016.
- J. Sindt, P. Camp, S. Kantorovich, E. Elfimova, and A. Ivanov. Influence of dipolar interactions on the magnetic susceptibility spectra of ferrofluids. Physical Review E, 93, 06 2016.
- A. O. Ivanov and P. J. Camp. Theory of the dynamic magnetic susceptibility of ferrofluids. Phys. Rev. E, 98:050602, Nov 2018.
- S. Kantorovich, E. Pyanzina, and F. Sciortino. The influence of shape anisotropy on the microstructure of magnetic dipolar particles. Soft Matter, 9:6594–6603, 2013.